

TEACHER'S NAME: _____
STUDENT'S NAME: _____

BAULKHAM HILLS HIGH SCHOOL

YEAR 11

YEAR 11 YEARLY EXAMINATION

2008

MATHEMATICS
EXTENSION 1

*Time allowed - 2 hours
(Plus five minutes reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- The 7 questions are of equal value. Start each question on a new page.
- All necessary working should be shown.
- Write your teacher's name and your name on the cover sheet provided.
- At the end of the exam, staple your answers in order behind the cover sheet provided.

QUESTION 1

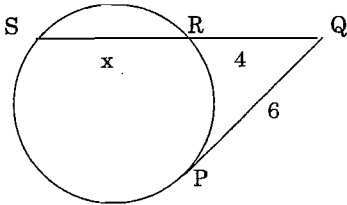
- (a) Calculate, to the nearest minute, the acute angle between the lines $x - 2y = 4$ and $3x + y = 5$. 3
- (b) Find the exact value of $\cos 15^\circ$ 2
- (c) Solve $\frac{2x+1}{x-2} \leq 1$ 3
- (d) Solve $9^x + 6(3^x) - 27 = 0$ 2
- (e) The quadratic equation $x^2 + 16x + k = 0$ has one root three times the other. Find k . 2

QUESTION 2 (Start a new page)

- (a) Prove that $\frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$. 3
- (b) Solve for $0^\circ \leq \theta \leq 360^\circ$
- (i) $\cos(2\theta - 30^\circ) = \frac{\sqrt{3}}{2}$ 2
- (ii) $7\cos^2 \theta - \sin \theta \cos \theta = 3$ 4
- (c) Find $f'(x)$ if $f(x) = (3x^2 - 2x)^{-1}$ 2
- (d) Evaluate $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 2x + 3}{x^2 + 2x + 1} \right)$ 1

QUESTION 3 (Start a new page)

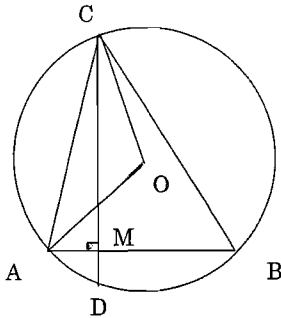
- (a) Simplify $\frac{4^n + 2^{n+3}}{2^{n-1}}$ 2
- (b) P(26,-16) divides the interval AB externally in the ratio 3:2. If A is (-4,-1), find the coordinates of B. 2
- (c) Find the number of terms in the series $e^x + e^{-x} + e^{-3x} + e^{-5x} + \dots + e^{-(2k+1)x}$. 3
- (d) PQ is a tangent to the circle. Find the value of x. 2



- (e) If the equation $x^2 - (k+1)x = 3k - 4$ has real roots, find the range of values of k. 3

QUESTION 4 (Start a new page)

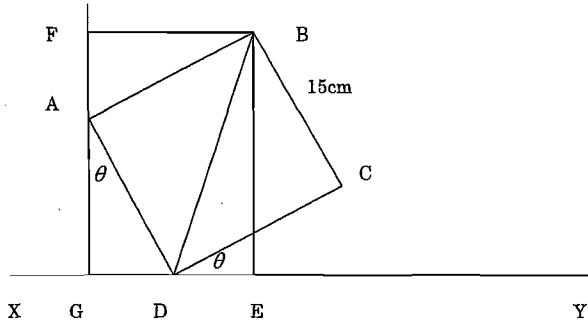
- (a) The limiting sum of a geometric series $1 + 3^x + 3^{2x} + 3^{3x} + \dots$ is $\frac{-1}{80}$. Find x. 2
- (b) Prove by mathematical induction that for $n \geq 1$, $11^n - 5^n$ is divisible by 6. 4
- (c) AB and CD are perpendicular chords of a circle, centre O. Prove $\angle DAB = \angle OAC$. 4



- (d) If $y = \frac{x-2}{x+2}$, prove that $y + \frac{dy}{dx} = \frac{x^2}{(x+2)^2}$. 2

QUESTION 5 (Start a new page)

- (a) In the diagram below ABCD is a square of side 15 cm leaning against a wall at an angle θ to the vertical and as well to the ground XY. 2



- (i) Show that $BD = 15\sqrt{2}$ cm 1
- (ii) Hence by using triangle DBE prove that the perpendicular distance from B to the line XY is $15\sqrt{2}\sin(45^\circ + \theta)$ 2
- (iii) By using triangles DAG and BFA find an expression for the length of FG. 2
- (iv) Hence prove that $\sin\theta + \cos\theta = \sqrt{2}\sin(45^\circ + \theta)$ 2
- (b) Find the equation of the parabola with focus (2,3) and directrix $y = -1$ 2
- (c) The elevation of a hill at a point P due east of it is 35° . At a point Q due south of P the elevation is 25° . If $PQ = 300$ m, find the height of the hill to the nearest metre. 3

QUESTION 6 (Start a new page)

- (a) Using first principles find $\frac{dy}{dx}$ if $y = 2\sqrt{x}$. 3
- (b) Consider the function $y = \frac{6}{3+x^2}$. 1
- (i) Show that the function is even. 1
- (ii) Find the coordinates of any stationary points and determine their nature. 3
- (iii) Sketch the curve. 2
- (iv) State the range of the function. 1
- (c) If $\frac{\sin(A+B)}{\sin(A-B)} = \frac{5}{3}$ show that $\tan A = 4\tan B$. 2

QUESTION 7 (Start a new page)

- (a) A parabola is represented by the equations $x = 2at$, $y = at^2$.
- (i) Find $\frac{dy}{dt}$, $\frac{dx}{dt}$ and hence find $\frac{dy}{dx}$. 1
- (ii) Hence find the equation of the normal to the parabola at $P(2ap, ap^2)$ 2
- (b) (i) If $t = \tan \frac{\alpha}{2}$, find expressions for $\sin \alpha$ and $\cos \alpha$ in terms of t . 2
- (ii) Using part (i) solve $\sqrt{3} \sin \alpha - \cos \alpha = 1$ for $0^\circ \leq \alpha \leq 360^\circ$ 3
- (c) Sam and Bob set out from two towns. They travel on roads that meet at right angles and they walk towards the intersection. Sam is initially 15 km from the intersection and walks at 3 km/h. Bob is initially 10 km from the intersection and walks at 4 km/h.
- (i) Show that their distance apart after t hours is given by $D^2 = 25t^2 - 170t + 325$. 1
- (ii) Hence find how long it takes them to reach their minimum distance apart. 3

END OF PAPER

Q1(a) $y = \frac{1}{2}x - 2$ $m_1 = \frac{1}{2}$
 $y = -3x + 5$ $m_2 = -3$ (1)

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$ (1)
 $= \left| \frac{\frac{1}{2} - (-3)}{1 + \frac{1}{2} \cdot (-3)} \right|$

$= \left| \frac{\frac{7}{2}}{1 - \frac{3}{2}} \right|$

$= 7$
 $\theta = 81^\circ 52'$ (1)

(b) $\cos 15^\circ = \cos(45^\circ - 30^\circ)$ (1)

$= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ$

$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$

$= \frac{\sqrt{3} + 1}{2\sqrt{2}}$

$\therefore = \frac{\sqrt{6} + \sqrt{2}}{4}$ (1)

(c) $\frac{2x+1}{x-2} \cdot (x-2)^2 \leq 1 \cdot (x-2)^2$ (1)

$(2x+1)(x-2) - (x-2)^2 \leq 0$

$(x-2)[2x+1 - (x-2)] \leq 0$

$(x-2)(x+3) \leq 0$ (1)

$-3 \leq x \leq 2$

But $x \neq 2$

$\therefore -3 \leq x < 2$ (1)

(d) $(3^x)^2 + 6 \cdot 3^x - 27 = 0$

$m^2 + 6m - 27 = 0$

$(m+9)(m-3) = 0$

$\therefore m = -9$ or $m = 3$

$3^x = -9$ $3^x = 3$

No soln $x = 1$ (2)

(e) let roots be $x, 3x$

$\therefore x + 3x = -\frac{b}{a} = -16$ (1)

$\therefore x = -4$

and $x \cdot 3x = \frac{c}{a} = k$

$3 \times 16 = k$

$k = 48$ (1)

Q2(a) LHS = $1 - \frac{(1 - 2\sin^2 \theta)}{2\sin \theta \cos \theta}$ (1)

$= \frac{2\sin \theta \cos \theta}{2\sin \theta \cos \theta}$

$= \frac{\sin \theta}{\cos \theta}$ (1)

$= \tan \theta = \text{RHS}$ (1)

(b)(i) $2\theta = 30^\circ, 30^\circ, 330^\circ, 330^\circ, 690^\circ$

$2\theta = 0^\circ, 60^\circ, 360^\circ, 420^\circ, 720^\circ$

$\theta = 0^\circ, 30^\circ, 180^\circ, 210^\circ, 360^\circ$ (1)

(ii) divide by $\cos^2 \theta$

$2 - \frac{\sin \theta}{\cos \theta} = \frac{3}{\cos^2 \theta}$

$2 - \tan \theta = 3 \sec^2 \theta$

$2 - \tan \theta = 3(1 + \tan^2 \theta)$ (1)

$3 \tan^2 \theta + \tan \theta - 4 = 0$

$3 \tan^2 \theta - 3 \tan \theta + 4 \tan \theta - 4 = 0$

$3 \tan \theta (\tan \theta - 1) + 4(\tan \theta - 1) = 0$

$(\tan \theta - 1)(3 \tan \theta + 4) = 0$ (1)

$\therefore \tan \theta = 1$ or $\tan \theta = -\frac{4}{3}$

$\theta = 45^\circ, 225^\circ, 126^\circ 52', 306^\circ 52'$ (2)

(c) $f(x) = (3x^2 - 2x)^{-1}$

$f'(x) = -(3x^2 - 2x)^{-2} \cdot (6x - 2)$ (2)

$m = \frac{2 - 6x}{(3x^2 - 2x)^2}$

(d) $\text{Exp} = \lim_{x \rightarrow \infty} \left(1 + \frac{2}{x} + \frac{3}{x^2} \right)$ (1)

$= \frac{1 + 0 + 0}{1 + 0 + 0}$

$= 1$

Q3(a) $\frac{2^{2n} + 2^{n+3}}{2^{n-1}} = \frac{2^{n-1}(2^{n+1} + 2^4)}{2^{n-1}}$ (1)
 $= 2^{n+1} + 2^4 = 2^{n+1} + 16$ (1)

(b) $x_p = \frac{mx_2 + nx_1}{m+n}$ (1) $y_p = \frac{my_2 + ny_1}{m+n}$

$26 = \frac{3x_2 + 2 \cdot 4}{3+2}$ $-16 = \frac{3y_2 + 2 \cdot 1}{3+2}$

$26 = 3x_2 + 8$ $-16 = 3y_2 + 2$

$3x_2 = 18$ $3y_2 = -18$

$x_2 = 6$ $y_2 = -6$

$\therefore B = (6, -6)$ (1)

(c) If internal (1) mark

G.S where $a = e^x$, $r = e^{-2x}$

$T_n = a r^{n-1}$ (1)

$e^{-(2k+1)x} = e^x (e^{-2x})^{n-1}$ (1)

$= e^x \cdot e^{-2nx+2x}$

$= e^{(-2n+3)x}$

$\therefore -2n+3 = -(2k+1)$

$2n = 2k+4$

$n = k+2$

$\therefore \text{No. of terms} = k+2$ (1)

(d) $(x+4) \cdot 4 = 6^2$ (1)

$4x = 20$

$x = 5$ (1)

(e) $x^2 - (k+1)x + 4 - 3k = 0$

If real roots $\Delta \geq 0$ (1)

$[-(k+1)]^2 - 4 \cdot 1 \cdot (4 - 3k) \geq 0$

$k^2 + 2k + 1 - 16 + 12k \geq 0$

$k^2 + 14k - 15 \geq 0$ (1)

$(k+15)(k-1) \geq 0$

$\therefore k \leq -15$ or $k \geq 1$ (1)

Q4 (a) $a = 1$, $r = 3^x$

$S_{\infty} = \frac{a}{1-r}$ (1)

$-80 = \frac{1}{1-3^x}$

$1-3^x = -80$

$3^x = 81$

$\therefore x = 4$ (1)

(b) i) Prove true for $n = 1$

$11^1 - 5^1 = 6 = 6 \times 1$

$\therefore$ True for $n = 1$ (1)

(ii) Assume true for $n = k$

i.e. $11^k - 5^k = 6P$ where $P \in \mathbb{Z}$ (1)

(iii) Prove true for $n = k+1$

i.e. $11^{k+1} - 5^{k+1} = 6Q$ where $Q \in \mathbb{Z}$

LHS = $11 \cdot 11^k - 5 \cdot 5^k$

$= 11(6P + 5^k) - 5 \cdot 5^k$ by assum.

$= 66P + 11 \cdot 5^k - 5 \cdot 5^k$

$= 66P + 6 \cdot 5^k$

$= 6(11P + 5^k)$ (1)

$= 6Q$ since $11P + 5^k$ is an integer

$\therefore$ True for $n = k+1$ if true for $n = k$.

(iv) Since true for $n = 1$ and since

true for $n = k+1$ if true for $n = k$

then it is true for $n = 2, 3, 4$, etc

$\therefore$ True for $n \geq 1$. (1)

(c) Draw AD let $\angle DAB = x^\circ$

$\therefore \angle DCB = x^\circ$ ($\angle$'s on circ. str on same arc) (1)

Let $\angle CBA = y^\circ$

$\therefore \angle ABC = 2y^\circ$ ($\angle$ at centre = $2 \times \angle$ at circ.)

ΔAOC is isos (2 eq. radii) circ.

$\therefore \angle OAC = \frac{1}{2}(180^\circ - 2y^\circ)$ ($\angle$ sum of Δ) (1)

$= 90^\circ - y^\circ$

But $x + y = 90^\circ$ ($\angle$ sum of right ΔBOC)

$\therefore \angle OAC = x^\circ$ (1)

(d) $\frac{dy}{dx} = \frac{(x+2) \cdot 1 - (x-2) \cdot 1}{(x+2)^2} = \frac{4}{(x+2)^2}$ (1)

$\therefore y + \frac{dy}{dx} = \frac{x-2}{x+2} + \frac{4}{(x+2)^2}$

$= \frac{x^2 - 4 + 4}{(x+2)^2}$ (1)

$= \frac{x^2}{(x+2)^2} = \text{RHS}$

$$56(x) BD^2 = BC^2 + DC^2 = 15^2 + 15^2 = 2 \times 15^2 \quad (1)$$

$$\therefore BD = 15\sqrt{2} \text{ cm}$$

$\angle BDC = 45^\circ$ (Diag of square bisect $\angle$)

$$\therefore \angle BDE = 45^\circ + \theta$$

$$\text{In } \triangle DBE, \frac{BE}{DB} = \sin(45^\circ + \theta) \quad (1)$$

$$BE = 15\sqrt{2} \sin(45^\circ + \theta) \quad (1)$$

In $\triangle FAB$, $\angle FAB = 90^\circ - \theta$ (sum of str. $\angle$)

$$\frac{FB}{AB} = \cos(90^\circ - \theta) \quad (1)$$

$$FB = 15 \cos(90^\circ - \theta) = 15 \sin \theta$$

$$\text{In } \triangle AGD, \frac{AG}{AD} = \cos \theta \therefore AG = 15 \cos \theta \quad (1)$$

$$\therefore FG = FB + AG = 15(\sin \theta + \cos \theta)$$

$$FG = BE = 15\sqrt{2} \sin(45^\circ + \theta) \quad (1)$$

$$\therefore 15(\sin \theta + \cos \theta) = 15\sqrt{2} \sin(45^\circ + \theta) \quad (1)$$

$$\therefore \sin \theta + \cos \theta = \sqrt{2} \sin(45^\circ + \theta)$$

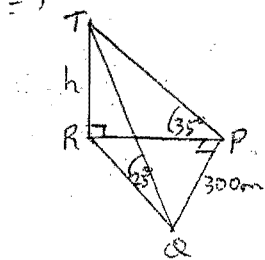
$$2a = 3 - 1 = 4 \therefore a = 2$$

$$\therefore V = (2, 1) \quad (1)$$

$$4 \sin^2(x - k) = 4a(y - k)$$

$$(x - 2)^2 = 8(y - 1) \quad (1)$$

$\Rightarrow$



$$\text{In } \triangle PRT, \frac{h}{RP} = \tan 35^\circ$$

$$RP = \frac{h}{\tan 35^\circ}$$

$$\text{In } \triangle QRT, \frac{h}{RQ} = \tan 15^\circ \quad (1)$$

$$RQ = \frac{h}{\tan 15^\circ}$$

$\therefore$ In $\triangle PQR$

$$RQ^2 = PQ^2 + PR^2$$

$$\frac{h^2}{\tan^2 15^\circ} = 300^2 + \frac{h^2}{\tan^2 35^\circ} \quad (1)$$

$$h^2 = \frac{300^2}{\frac{1}{\tan^2 15^\circ} - \frac{1}{\tan^2 35^\circ}}$$

$$= 35165.82166$$

$$h = 187.5 \div 188 \text{ m} \quad (1)$$

$$Q6(a) \text{ let } f(x) = 2\sqrt{x}$$

$$\therefore f(x+h) = 2\sqrt{x+h}$$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (1)$$

$$= \lim_{h \rightarrow 0} \frac{2\sqrt{x+h} - 2\sqrt{x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2\sqrt{x+h} - 2\sqrt{x}}{h} \cdot \frac{2\sqrt{x+h} + 2\sqrt{x}}{2\sqrt{x+h} + 2\sqrt{x}}$$

$$= \lim_{h \rightarrow 0} \frac{4(x+h) - 4x}{h(2\sqrt{x+h} + 2\sqrt{x})} \quad (1)$$

$$= \lim_{h \rightarrow 0} \frac{4}{2\sqrt{x+h} + 2\sqrt{x}}$$

$$= \frac{4}{4\sqrt{x}} = \frac{1}{\sqrt{x}} \quad (1)$$

$$(b)(i) f(x) = \frac{6}{3+x^2}$$

$$f(-x) = \frac{6}{3+(-x)^2} = \frac{6}{3+x^2} = f(x)$$

$\therefore$ Even

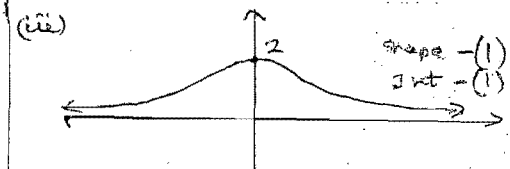
$$(ii) \frac{dy}{dx} = \frac{0 - 6 \cdot 2x}{(3+x^2)^2} = \frac{-12x}{(3+x^2)^2} \quad (1)$$

st pts when $\frac{dy}{dx} = 0$

$$\therefore x = 0, y = 2 \quad (1)$$

when $x < 0$, $\frac{dy}{dx} > 0$ when $x > 0$, $\frac{dy}{dx} < 0$

$\therefore$ Max. turn. pt at $(0, 2)$ (1)



(iv) Range: $0 < y \leq 2$ (1)

$$(c) 3 \sin A \cos B + 3 \cos A \sin B$$

$$= 5 \sin A \cos B - 5 \cos A \sin B \quad (1)$$

$$2 \sin A \cos B = 8 \cos A \sin B$$

$$\frac{\sin A}{\cos A} = 4 \frac{\sin B}{\cos B} \quad (1)$$

$$\tan A = 4 \tan B$$

$$Q7(a)(i) \frac{dy}{dt} = 2at \quad \frac{dx}{dt} = 2a$$

$$\therefore \frac{dy}{dx} = 2at \times \frac{1}{2a} = t \quad (1)$$

$$(ii) m_T = p \therefore m_N = -\frac{1}{p} \quad (1)$$

$$\therefore \text{Eqn is } y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$\text{or } x + py = 2ap + ap^3$$

$$(b)(i) \tan \frac{\alpha}{2} = \frac{t}{1}$$

$$\therefore \sin \frac{\alpha}{2} = \frac{t}{\sqrt{1+t^2}}$$

$$\cos \frac{\alpha}{2} = \frac{1}{\sqrt{1+t^2}}$$

$$\sin \alpha = 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2}$$

$$= 2 \frac{t}{\sqrt{1+t^2}} \cdot \frac{1}{\sqrt{1+t^2}}$$

$$\therefore \sin \alpha = \frac{2t}{1+t^2} \quad (1)$$

$$\cos \alpha = \cos^2 \frac{\alpha}{2} - \sin^2 \frac{\alpha}{2}$$

$$= \frac{1}{1+t^2} - \frac{t^2}{1+t^2}$$

$$\therefore \cos \alpha = \frac{1-t^2}{1+t^2} \quad (1)$$

$$(ii) \sqrt{3} \cdot \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2} = 1$$

$$2\sqrt{3}t - 1 + t^2 = 1 + t^2$$

$$2\sqrt{3}t = 2$$

$$\tan \frac{\alpha}{2} = \frac{1}{\sqrt{3}} \quad (1)$$

$$\therefore \frac{\alpha}{2} = 30^\circ$$

$$\alpha = 60^\circ \quad (1)$$

$$\text{Check } 180^\circ: \text{LHS} = \sqrt{3} \cdot 0 - 1 = -1 = \text{RHS}$$

$$\therefore \alpha = 60^\circ, 180^\circ \quad (1)$$

$$(c) S = 15 - 3t, B = 10 - 4t \quad (1)$$

$$D^2 = S^2 + B^2 = (15 - 3t)^2 + (10 - 4t)^2$$

$$= 225 - 90t + 9t^2 + 100 - 80t + 16t^2$$

$$= 25t^2 - 170t + 325$$

$$\frac{d(D^2)}{dt} = 50t - 170 \quad (1)$$

$$= 0 \text{ when } t = 3.4$$

$$\frac{d^2(D^2)}{dt^2} = 50 > 0 \therefore \text{Min distance}$$

$$\text{occur when } t = 3.4 \text{ h or } 31.2 \text{ km}$$